江苏省苏州中学 国际部
Suzhou High School – International Division

SZZX-ID
BP Physics

Force on a current carrying wire

Class Outline

Title of Unit	Introduction to magnetic field
Title of Lesson	Force on a current carrying wire
Starter Activity	Kahoot: Magnetism and magnetic field
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- demonstration: force between wires - lecture: left-hand rule - inquiry: worksheet
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	- work on worksheet as pairs - Group discussion for the last questions (challenge yourselves)
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: - Understand the strength and direction of magnetic force on a current-carrying wire - Understand and predict force between two wires - Apply left-hand rule to determine the direction of the force - Understand and explain otor effect – speaker and motor
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Thinking: Being curious about the natural world (how things work: motor and speaker)
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Inquirer, thinker
Teaching Activities	- demonstration: video: force on a wire; force between two wires - worksheet: left hand rule
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.

Magnetic Field Strength, B

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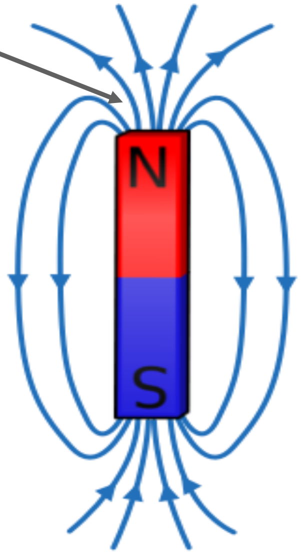
The magnetic field strength, B, can be thought of as the density of magnetic field lines per square meter (also called Magnetic Flux Density, B).

It is a vector.

It has units of Tesla, T.

The density of flux lines is great, therefore B is large.



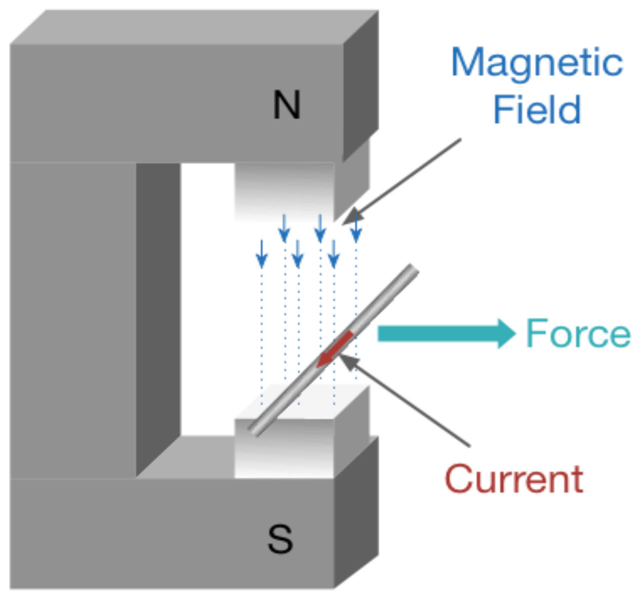
The density of flux lines is small, therefore B is small.

The Magnetic Force on Current

- **If a current-carrying wire is placed in a region of magnetic field, it will experience a magnetic force.**

- **The magnitude of force is maximized when the current is perpendicular to the magnetic field.**

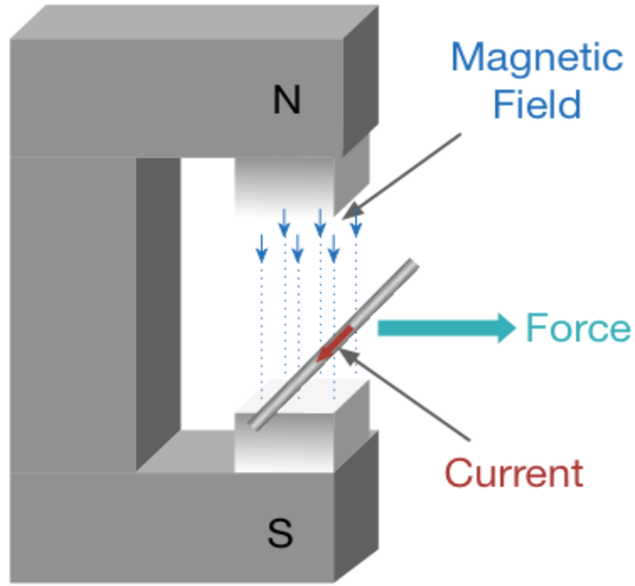
- **The force is 0 when the current is parallel to the magnetic field.**



The Magnetic Force on Current

To increase the magnitude of the force

- **Increase the current in the wire.**
- **Increase the strength of the magnetic field.**
- **Increase the length of the wire within the magnetic field.**
- **Increase the number of wires**



Determine the direction of magnetic force on a straight wire

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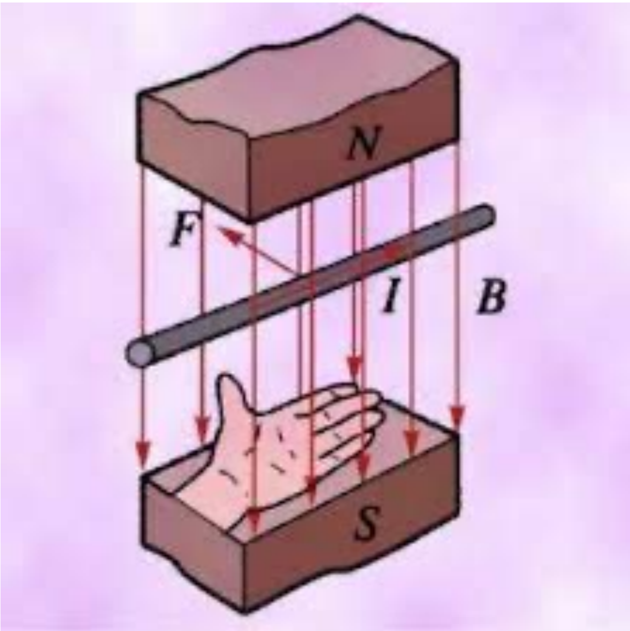
Left-hand rule



- Use **left hand**
- Let **magnetic field lines go into the palm**
- The **four fingers pointing to the direction of current**
- The **direction of the thumb is the direction of magnetic force** acting on the wire

Determine the direction of magnetic force on a straight wire – another way

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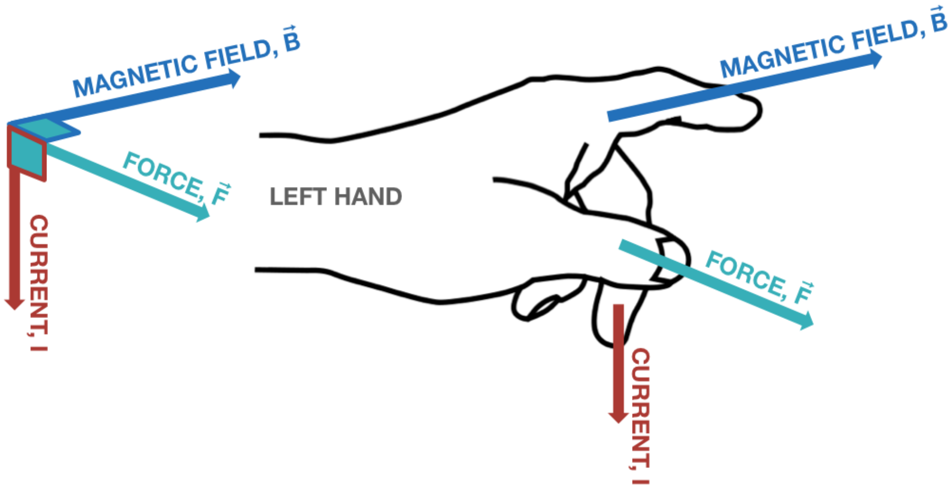
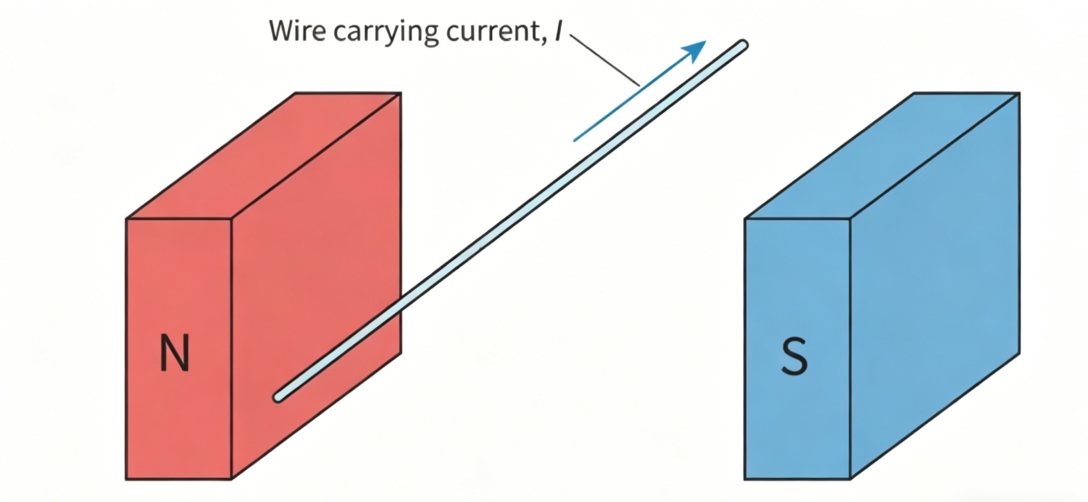
Fleming's Left Hand Rule

The direction of the magnetic force can be determined using Fleming's Left Hand Rule.



All three fingers should be mutually perpendicular.

Practice – left hand rule

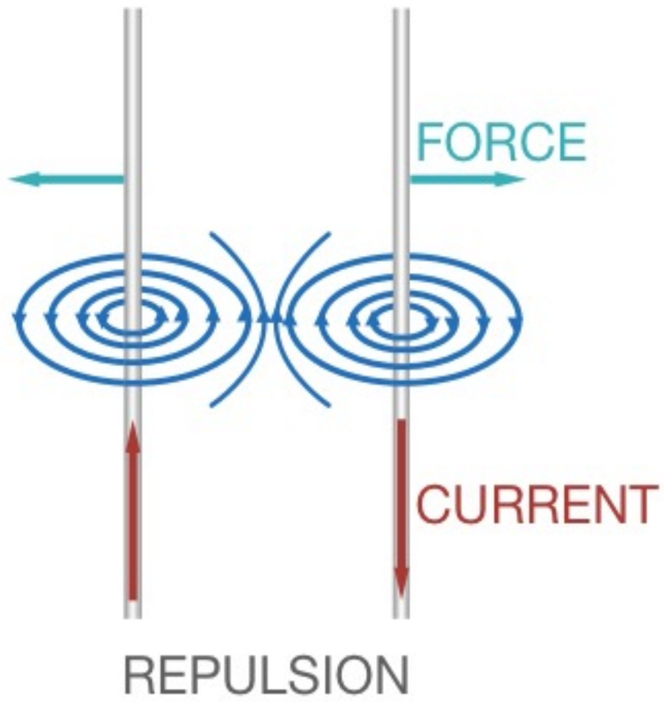


Choose your preferred left-hand rule and determine the direction of force.

Remember they are just tools. You just need to pick one and get familiar with it

- Field: left to right
- Current: into the page
- Force: down
- If either the current or the field are reversed, the force will be upwards.

Magnetic Force Between Two Wires

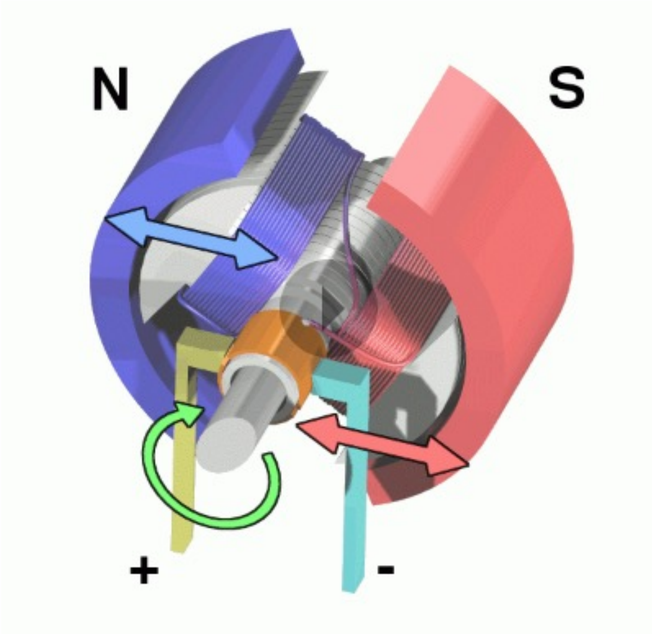


Two current-carrying wires brought close to each other will experience a force.

- Opposing currents produce a repulsive force.
- Like currents produce an attractive force.

The D.C. motor

- Electric motors are incredibly reliable. They only have one moving part and can spin very slowly or very quickly without the use of gears.
- Applications include lifts, washing machines, trains, blenders, robots, drones and even cars.



Source: 'Electric motor' by Abnormaal, CC-BY-SA-3.0

The D.C. motor

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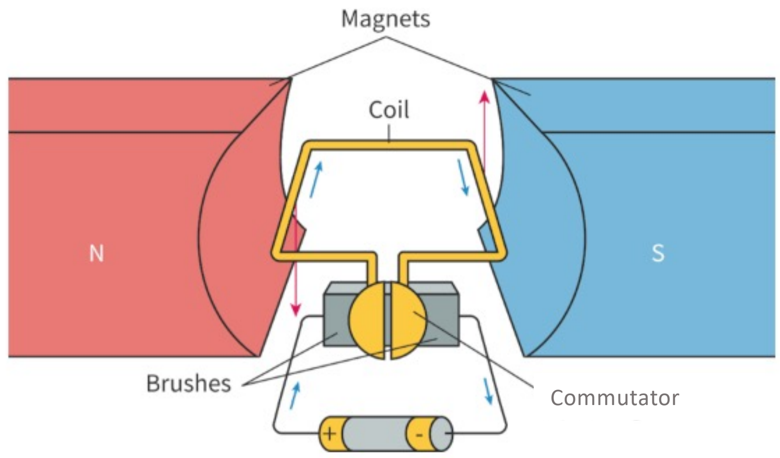
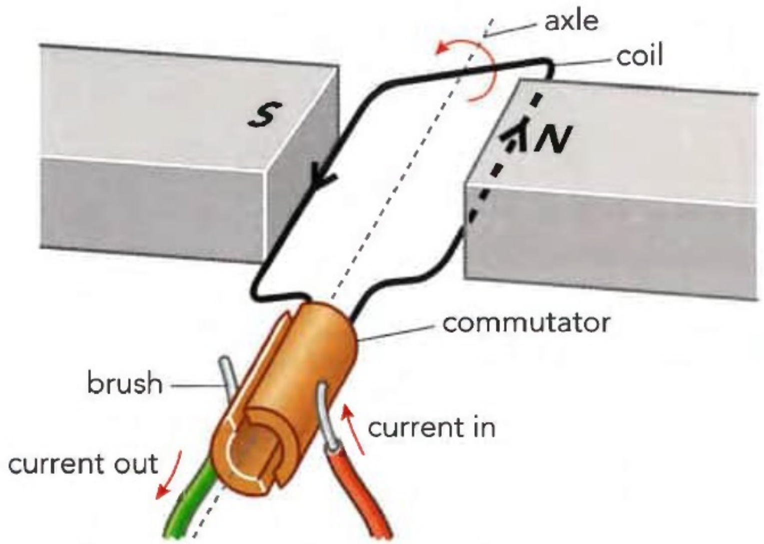
- The left-hand side of the coil is perpendicular to the magnetic field and therefore experiences a force downwards.
- The right arm of the coil is carrying current in the opposite direction and is also perpendicular to the magnetic field, and so experiences a force upwards.
- A turning force causes the coil to spin about the pivot.



To increase the turning force of a motor, you can:
- increase the current
- increase the strength of the magnetic field
- increase the number of turns in the coil.

Keeping the motor turning



Each half-turn, the commutator reverses the connections to the power supply, so that the motor keeps turning in the sme direction

The loudspeaker

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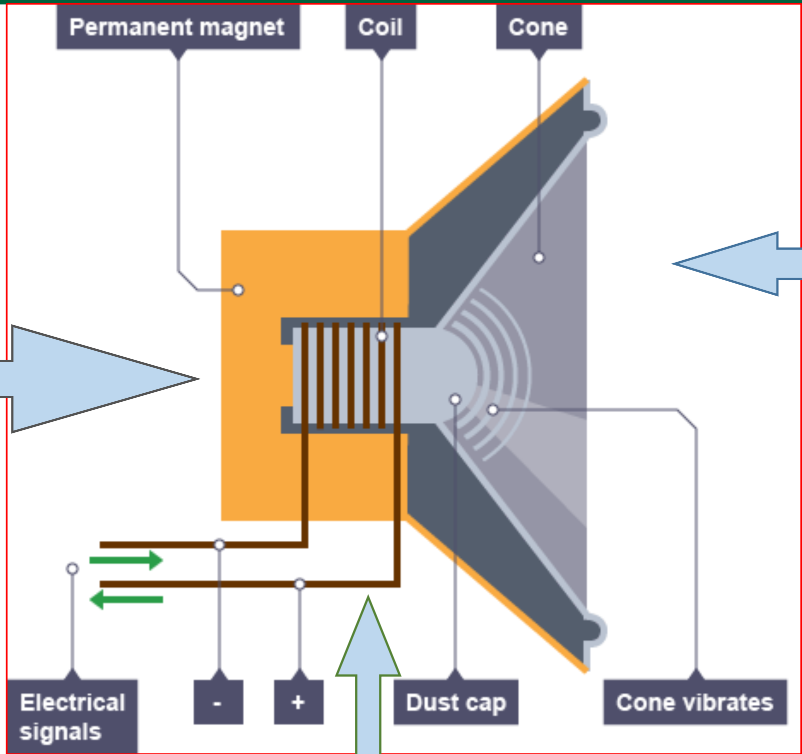
1. The **cylindrical magnet** produces a strong **radial** magnetic field at **right angles** to the wire in the coil, which is **attached** to a paper or plastic cone.



3. The speaker cone also moves back and forth in response to the electrical signal, generating sound waves.

2. When a current flows in the wire, the wire and cone will move. The current input. Caries with the sound being produced- a varying signal received from a sound processing device (e.g. a computer)